Examiner only

4. Port Talbot steel works in South Wales is one of the largest in the UK. It produces 5 million tonnes of steel every year.

The materials required for the production of one tonne of iron are shown in the table below. The melting point of iron is 1535 °C.

(a) Describe the purpose of the following raw materials in the blast furnace. [2]

Haematite ....

Limestone ....

(b) The blast furnace relies on the processes of reduction and oxidation. In one stage carbon monoxide is formed from coke.

(i) **Complete** the balanced symbol equation for the extraction of iron in the blast furnace. [3]

Fe_2O_3 + ... $\longrightarrow$ .... + ...

(ii) State the substances that are reduced or oxidised in the reactions that must take place to produce iron in the blast furnace. [2]

....

....

....

(c) The materials required for the production of one tonne of iron are shown in the table below. The melting point of iron is 1535 °C.

Raw material	Mass of raw material (tonnes)	Cost per tonne of raw material (£)
haematite (iron ore)	1.75	60.50
coke	0.25	120.90
limestone	0.25	80.00
air	4.00	2.25

Calculate the **monthly** raw material cost, for the Port Talbot steel works. Give your answer to two significant figures. [4]

= £ .... million / month

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